

ABSTRACT

A sensitive method for quantification of multiple *Leishmania* species in mammalian cells and *in vivo* infections

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Leishmaniasis is a neglected disease caused by an obligate intracellular protozoa of the genus *Leishmania*. The disease has different clinical manifestations, such as cutaneous, mucocutaneous and visceral leishmaniasis that depends on the infecting *Leishmania* species, as well as on genetics factor and the host's immune response. According to the WHO and CDC, the detection of parasite DNA through qPCR improves the sensitivity and specificity of the diagnosis, and also enables parasite quantification. Thus, an efficient and straightforward method that detect and monitors parasite load is crucial for disease control. In this study we describe a standardized method based on qPCR for the detection and quantification of different *Leishmania* species in infected mammalian host cells. We were able to identify the *L. braziliensis* and *L. guyanensis* species using TaqMan technology, while SYBR Green was able to classify three groups of *Leishmania*. Our method also allows quantify parasites in several types of samples from different tissues of mammalian hosts. The development of a sensitive method based on qPCR for parasite load quantification can greatly improve the treatment and control of Leishmaniasis.

Keywords: Population Genetics; Leishmaniasis; Parasite Load.

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